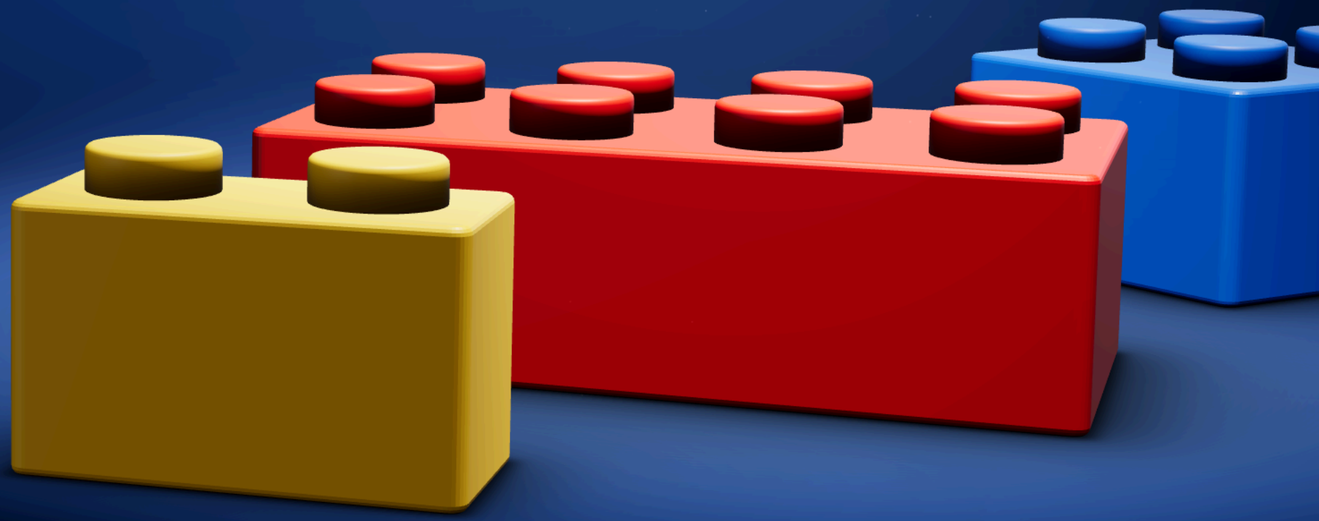
Are LEGO themes real universes?

A few are. Most are not: they are weak universes, built from the same bricks.

9,062 sets | **317,495** links | **1990–2025**



Enzo · Keira · Amandine

Graph Analytics, Albert School · Rebrickable data

Many theme-specific parts, but the same plates everywhere

77%

of the 28,856 parts appear in only one theme. At the other end, 51 parts are used in 100 themes or more.

A few themes have their own parts:

Technic	3×2 seat, linear actuator, mudguard
Duplo	a different scale: stool, steak
Super Mario	Warp Pipe, Goomba tile

Star Wars, Friends: mostly printed standard parts and accessories.



Plate 1×2
in 73% of sets



Plate 2×4
in 65% of sets

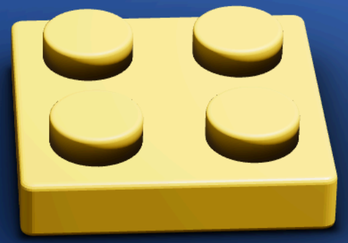


Plate 2×2
in 63% of sets



Plate 1×4
in 63% of sets

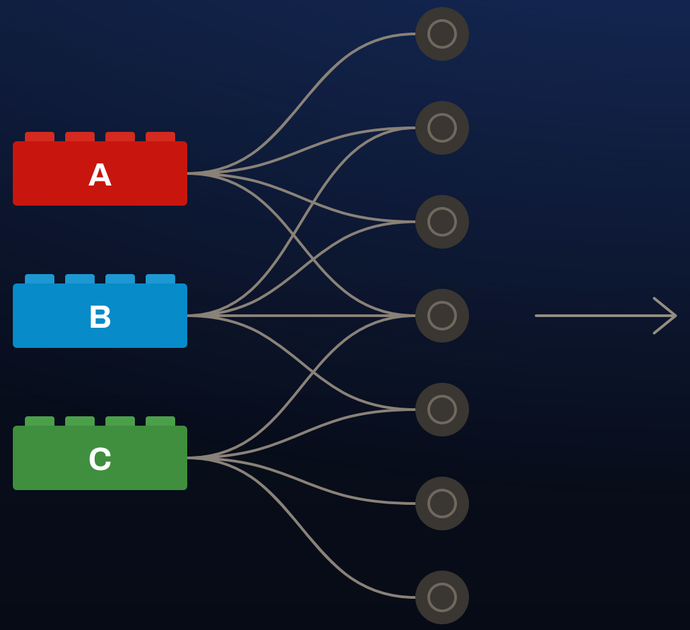
results/parts_universality.csv · signature_parts.csv

Two sets are linked if they share enough parts

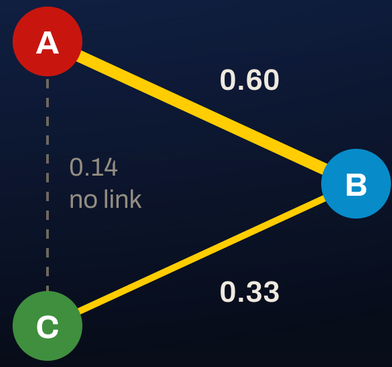
- Rebrickable part inventories
- Sets from 1990–2025, at least 50 parts
- Excluding gear, books, key chains, bulk bricks and collectible minifigures
- Subthemes merged into their theme: Police → City
- One part = one shape, color ignored

9,062 sets · **317,495** links

Example: 3 sets, 7 parts



Set graph

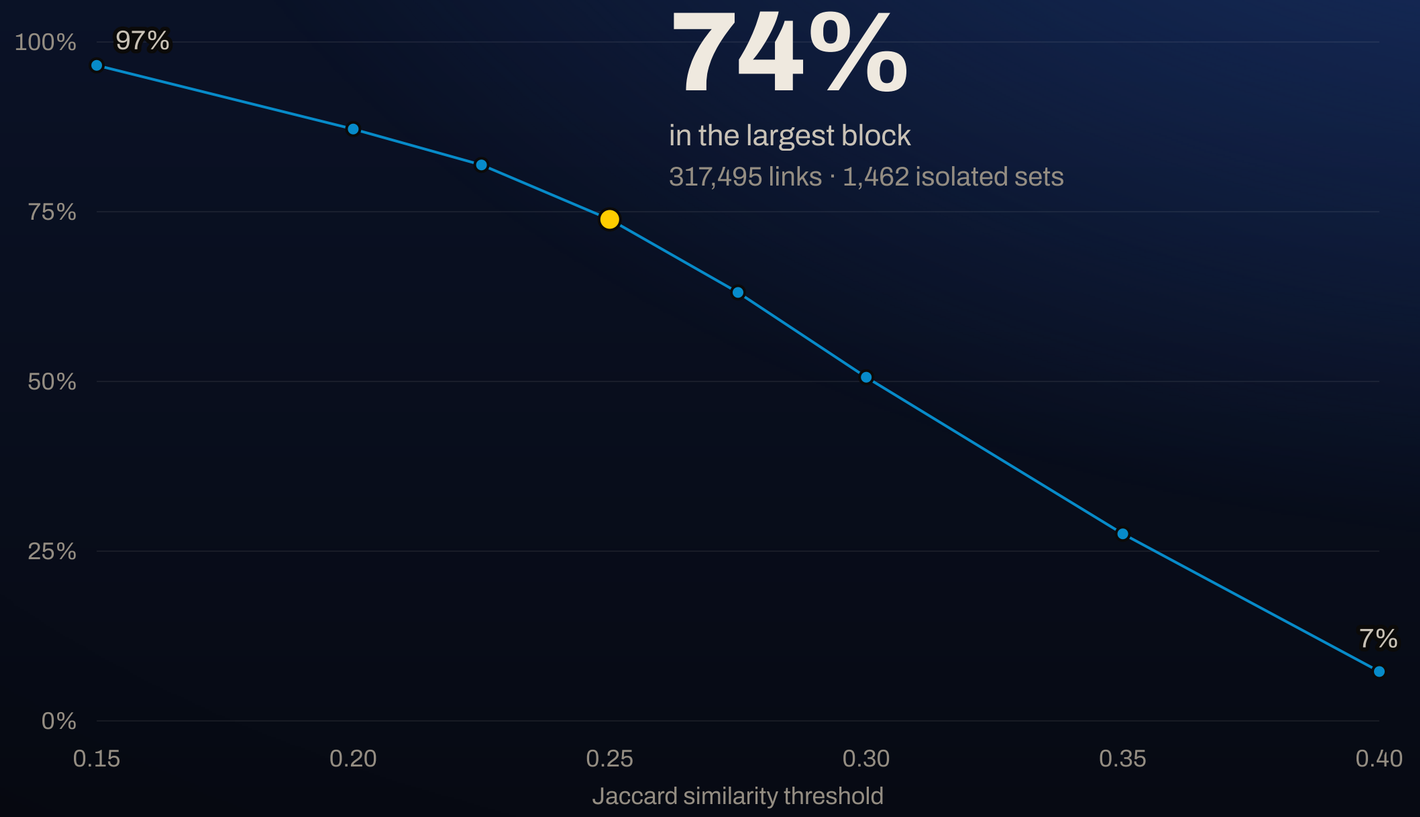


Jaccard(A, B) = shared parts ÷ parts in A or B

Link if J ≥ 0.25. Here A–B: 3 ÷ 5 = 0.60

At 0.25, 74% of sets stay in a single block

Share of sets in the largest block



We tested 8 thresholds, from 0.15 to 0.40.

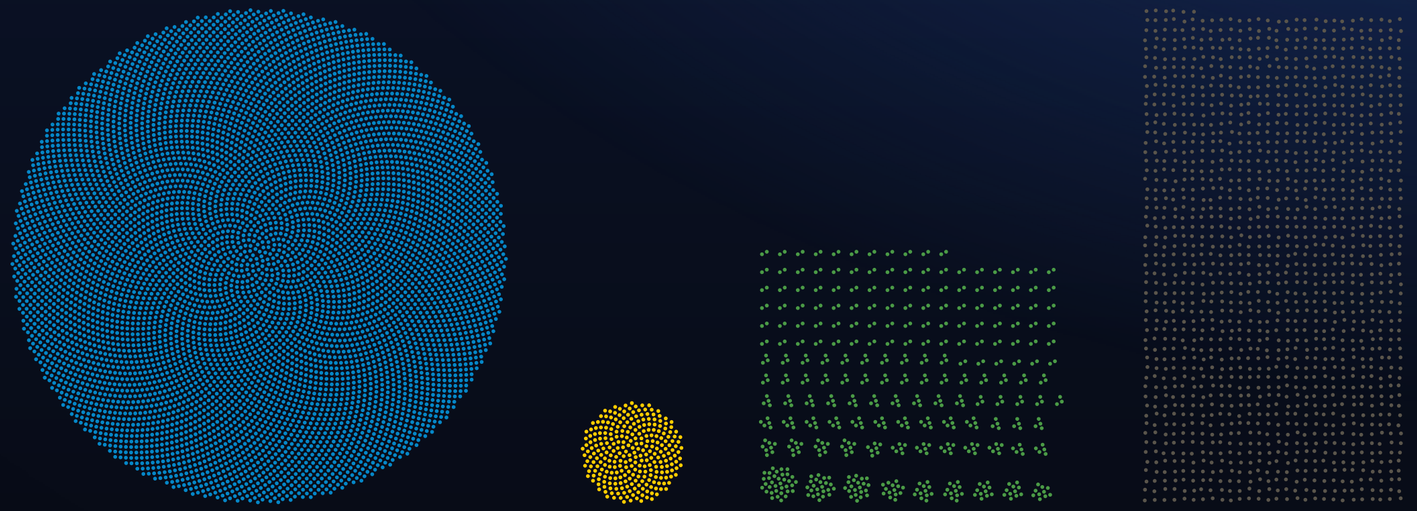
Lower, almost everything connects: 4.6 million links at 0.15.

Higher, the graph falls apart: only 7% of sets in the largest block at 0.40.

Only results that hold across several thresholds are presented as conclusions.

results/sensitivity.csv

One large block that mixes almost every theme, with Duplo on its own



6,694 sets
the largest block (74%)

282 sets
including 216 Duplo

624 sets
176 small groups

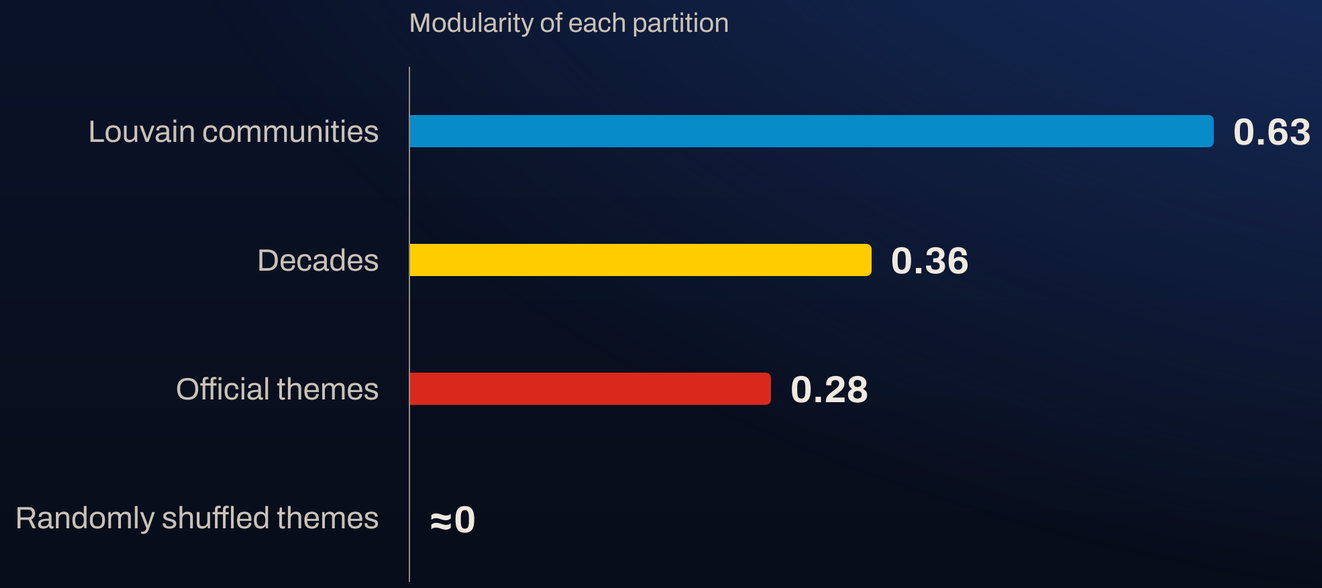
1,462 sets
isolated, with no links

Density	**0.77%**
Mean degree	**70** neighbors
Diameter	**16** steps
Mean distance (estimated)	**3.8** steps

The large block mixes 117 of the 127 themes.

The most connected sets (The Brick Moulding Machine, R2-D2) are not the bridges between groups (Architecture Studio, Fallingwater).

Era shapes the graph more than theme does



Modularity: share of links that stay inside groups, minus what chance alone would give (0 = no better than chance).

0.45

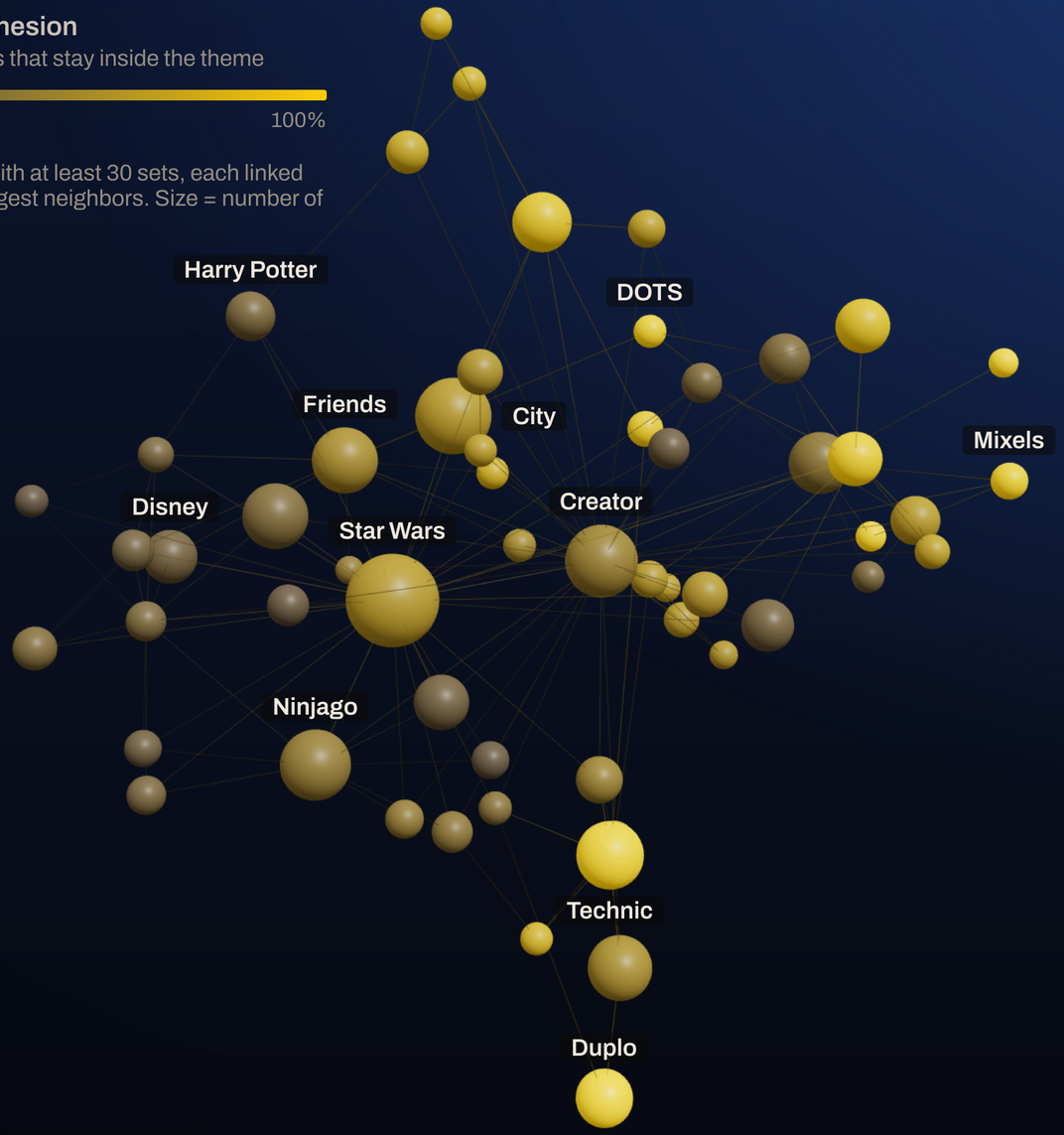
NMI between Louvain communities and official themes, non-isolated sets (1 = identical, 0 = unrelated).

Louvain at 4 resolutions and 2 label propagation variants: NMI between 0.43 and 0.54.

Theme cohesion
share of links that stay inside the theme

0% 100%

57 themes with at least 30 sets, each linked to its 3 strongest neighbors. Size = number of sets.



Neighbors give away a set's theme

Neighbor vote **63.9%**

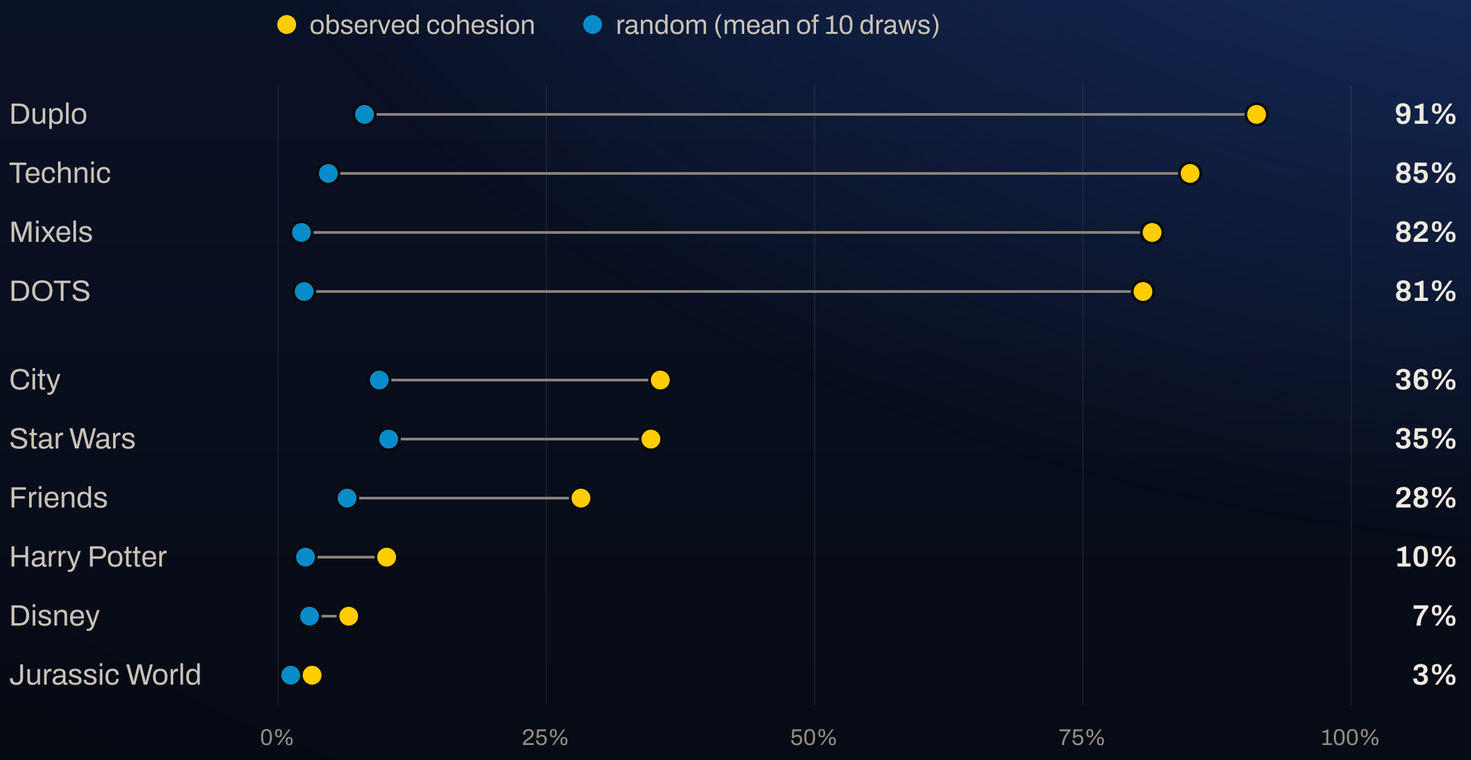
Always the largest theme **8.4%**

Random guess **3.1%**

Share of sets whose theme is recovered by a weighted vote of their neighbors, excluding the set itself (non-isolated sets).

results/theme_graph_edges.csv · extensions_summary.json

A few closed universes, many weak ones



6 of the 57 themes exceed 80%: the top 4, plus Clikits and Belville.

Cohesion
share of the theme's links that stay inside the theme

Random
parts are shuffled between sets, keeping each set's size and each part's frequency

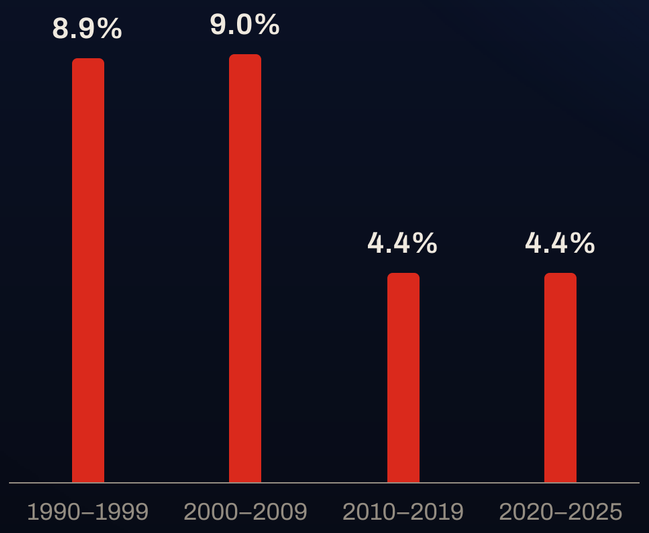
Every theme (57 of 57) is more cohesive than in each of the 10 draws.

Licensed themes: 2 to 4 times more than random, far from the closed universes.

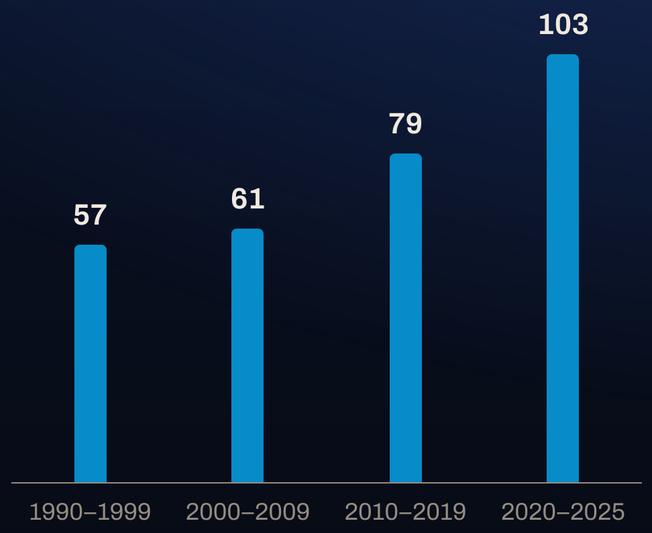
results/theme_cohesion_null.csv

No specialization: sets grow with common parts

Share of theme-exclusive parts



Distinct parts per set



Same trend without stickers: 8.3% → 4.0%. This measure does not depend on the threshold.

Limitations

The 0.25 threshold is a choice that changes the graph a lot.

Printed parts and stickers count as separate parts.

Null model: only 10 draws.

Part color is ignored.

results/temporal.csv · temporal_robustness.csv

A few themes are real universes; most are weak universes, built from the same bricks.

Closed universes	6 of 57 themes keep over 80% of their links, including Duplo and Technic.
Weak universes	licensed themes, only 2 to 4 times more cohesive than random.
No specialization	theme-exclusive parts: 8.9% in the 1990s, 4.4% since 2020.

